Problem 1: Mann–Whitney U Test

Dataset: `PlantGrowth` (built-in R dataset)

A botanist investigates whether a new fertilizer (`trt1`) affects plant growth compared with plants grown under standard conditions (`ctrl`).
The response variable is the dry weight of each plant (grams).

Tasks

1. Load the dataset `PlantGrowth`, Select only the observations from the `ctrl` and `trt1` groups. Construct side-by-side boxplots of plant weights for the two groups.
2. State the null and alternative hypotheses for comparing the two independent samples.
3. Perform the Mann–Whitney U (Wilcoxon rank-sum) test at the significance level $\alpha = 0.05$.
4. Compare your conclusion with that obtained from the two-sample Student's t-test. Are the conclusions the same?
5. Explain why the Mann–Whitney U test may be preferred to the t-test in this study.

Problem 2: Wilcoxon Signed-Rank Test

Dataset: `sleep` (built-in R dataset)

A clinical study investigates whether two different drugs produce different increases in sleep duration. Each subject receives both drugs, making the observations paired. The variable `extra` records the increase in hours of sleep, while `group` indicates the drug administered.

Tasks

1. Load the dataset `sleep`, explain why these data should be analyzed as paired samples, construct appropriate graphical summaries (boxplots or paired plots).
2. State the null and alternative hypotheses.
3. Perform the Wilcoxon Signed-Rank test using a significance level $\alpha = 0.05$. State your statistical decision. Interpret the result in terms of the effectiveness of the two drugs.
4. Perform the paired Student's t-test. Compare the conclusions obtained from the Wilcoxon Signed-Rank test and the paired t-test.

Problem 3: Kruskal–Wallis Test

Dataset: `iris` (built-in R dataset)

A botanist wants to determine whether the distribution of petal lengths differs among the three iris species: *Setosa*, *Versicolor*, and *Virginica*.

Tasks

1. Load the dataset `iris` into R. Construct side-by-side boxplots of Petal Length for the three species. Compute descriptive statistics for each species.
2. State the null and alternative hypotheses. Perform the Kruskal–Wallis test at the significance level $\alpha = 0.05$. State your statistical decision. Interpret the result in the context of the biological experiment.
3. If the null hypothesis is rejected, perform pairwise Wilcoxon rank-sum tests using the Bonferroni correction. Identify which pairs of iris species differ significantly.

Problem 4: Bootstrap Confidence Interval for the Median

Dataset: `ToothGrowth` (built-in R dataset)

A researcher investigates the effect of vitamin C on tooth growth in guinea pigs. Consider only the animals that received a dose of **1 mg/day** of vitamin C.
The objective is to estimate the **population median tooth length** and construct a **95% bootstrap confidence interval** for the median.
Tasks

- Load the dataset `ToothGrowth`, select only the observations corresponding to dose $= 1$.
- Construct a histogram and or a boxplot of tooth length, compute some descriptive statistics.
- Explain why the bootstrap method is appropriate for estimating the confidence interval of the median.

Problem 4: Bootstrap Confidence Interval for the Median

- Generate $B = 5000$ bootstrap samples by sampling with replacement. Compute the sample median for each bootstrap sample. Construct the bootstrap distribution of the median using a histogram. Estimate the bootstrap standard error of the median.
- Construct a 95% bootstrap confidence interval for the population median using the percentile method. Interpret the confidence interval in the context of the study.
- Repeat the analysis using the `boot` package and compare the resulting confidence interval with the one obtained from your own bootstrap algorithm.